(iii) $\tan 3A - \tan 2A - \tan A = \tan 3A + \tan(-2A) + \tan(-A) = \tan 3A \tan 2A \tan A$

(iv) If $A + B + C = \frac{\pi}{2}$, then $\tan A \tan B + \tan B \tan C + \tan C \tan A = 1$ (How!).

Exercise - 3.4

1. If $\sin x = \frac{15}{17}$ and $\cos y = \frac{12}{13}, 0 < x < \frac{\pi}{2}, 0 < y < \frac{\pi}{2}$,
   find the value of (i) $\sin(x + y)$ (ii) $\cos(x - y)$ (iii) $\tan(x + y)$.
2. If $\sin A = \frac{3}{5}$ and $\cos B = \frac{9}{41}, 0 < A < \frac{\pi}{2}, 0 < B < \frac{\pi}{2}$,
   find the value of (i) $\sin(A + B)$ (ii) $\cos(A - B)$.
3. Find $\cos(x - y)$, given that $\cos x = -\frac{4}{5}$ with $\pi < x < \frac{3\pi}{2}$ and $\sin y = -\frac{24}{25}$ with $\pi < y < \frac{3\pi}{2}$.
4. Find $\sin(x - y)$, given that $\sin x = \frac{8}{17}$ with $0 < x < \frac{\pi}{2}$ and $\cos y = -\frac{24}{25}$ with $\pi < y < \frac{3\pi}{2}$.
5. Find the value of (i) $\cos 105^\circ$ (ii) $\sin 105^\circ$ (iii) $\tan \frac{7\pi}{12}$.
6. Prove that (i) $\cos(30^\circ + x) = \frac{\sqrt{3}\cos x - \sin x}{2}$ (ii) $\cos(\pi + \theta) = -\cos\theta$
   (iii) $\sin(\pi + \theta) = -\sin\theta$.
7. Find a quadratic equation whose roots are $\sin 15^\circ$ and $\cos 15^\circ$.
8. Expand $\cos(A + B + C)$. Hence prove that
   $\cos A \cos B \cos C = \sin A \sin B \cos C + \sin B \sin C \cos A + \sin C \sin A \cos B$,
   $if\ A + B + C = \frac{\pi}{2}$.
9. Prove that
   (i) $\sin(45^\circ + \theta) - \sin(45^\circ - \theta) = \sqrt{2}\sin\theta$.
   (ii) $\sin(30^\circ + \theta) + \cos(60^\circ + \theta) = \cos\theta$.
10. If $a\cos(x + y) = b\cos(x - y)$, show that $(a + b)\tan x = (a - b)\cot y$.
11. Prove that $\sin 105^\circ + \cos 105^\circ = \cos 45^\circ$.
12. Prove that $\sin 75^\circ - \sin 15^\circ = \cos 105^\circ + \cos 15^\circ$.
13. Show that $\tan 75^\circ + \cot 75^\circ = 4$.
14. Prove that $\cos(A + B)\cos C - \cos(B + C)\cos A = \sin B \sin(C - A)$.
15. Prove that $\sin(n + 1)\theta \sin(n - 1)\theta + \cos(n + 1)\theta \cos(n - 1)\theta = \cos 2\theta,\ n \in \mathbb{Z}$.
16. If $x\cos\theta = y\cos\left(\theta + \frac{2\pi}{3}\right) = z\cos\left(\theta + \frac{4\pi}{3}\right)$, find the value of $xy + yz + zx$.
17. Prove that
   (i) $\sin(A + B)\sin(A - B) = \sin^2 A - \sin^2 B$
   (ii) $\cos(A + B)\cos(A - B) = \cos^2 A - \sin^2 B = \cos^2 B - \sin^2 A$
   (iii) $\sin^2(A + B) - \sin^2(A - B) = \sin 2A \sin 2B$
   (iv) $\cos 8\theta \cos 2\theta = \cos^2 5\theta - \sin^2 3\theta$
18. Show that $\cos^2 A + \cos^2 B - 2\cos A \cos B \cos(A + B) = \sin^2(A + B)$.
19. If $\cos(\alpha - \beta) + \cos(\beta - \gamma) + \cos(\gamma - \alpha) = -\frac{3}{2}$, then prove that
   $\cos\alpha + \cos\beta + \cos\gamma = \sin\alpha + \sin\beta + \sin\gamma = 0$.